# Run report: ML-aided SAT ceiling for the PDP (run1)

**Study:** `research/pdp_ml_sat_ceiling_20261027/` · **Protocol:** PROTOCOL.md (commit 8a85584fd, amendment 1) · **Probe:** `examples/pdp_ml_sat_ceiling.rs`, seed 20261027.
**Labelling run:** started 2026-10-27T06:44:54Z at commit 7dddd8099 (`run1/rows_*.jsonl`). **Analysis run:** 2026-10-27T08:46:02Z at commit d673109e7, `--from-rows run1` (`run1/analysis/`), wall 8 min 11 s.
**Host:** local development container, 4 vCPU Intel Xeon @ 2.10 GHz, **no isolation** (not Iso Lab: no AWS identity). Cost unit: solver conflicts (machine-independent); wall times diagnostic only.
**Arms:** symmetrised $S_4$, $m = 3$, native CDCL+XOR, symmetry breaking on; families n15l5 (3000 targets) and n17l6 (1200); L3 on 60 / 30 held-out UNSAT targets, cap 10× a0 median.

**Runs requested and completed.** (1) First launch at commit d3d27b879: aborted by its own label cross-check — a solver soundness bug (fixed, 7dddd8099). (2) Labelling run: both families labelled; killed by the session's 2 h background limit during n17l6 L3; L1–L3 results were never written. (3) Analysis run from the saved rows: complete, exit 0. Every label (4200) agrees with exhaustive sorted-triple search.

**Population (measured).**

| family | n | l | targets | SAT | SAT conflicts median [IQR] | UNSAT conflicts median [IQR] | UNSAT min–max |
|---|---|---|---|---|---|---|---|
| n15l5 | 15 | 5 | 3000 | 576 | 2678 [1388, 4273] | 5781 [5738, 5819] | 5546–5941 |
| n17l6 | 17 | 6 | 1200 | 336 | 20229 [10324, 34124] | 45649 [45539, 45781] | 44924–46297 |

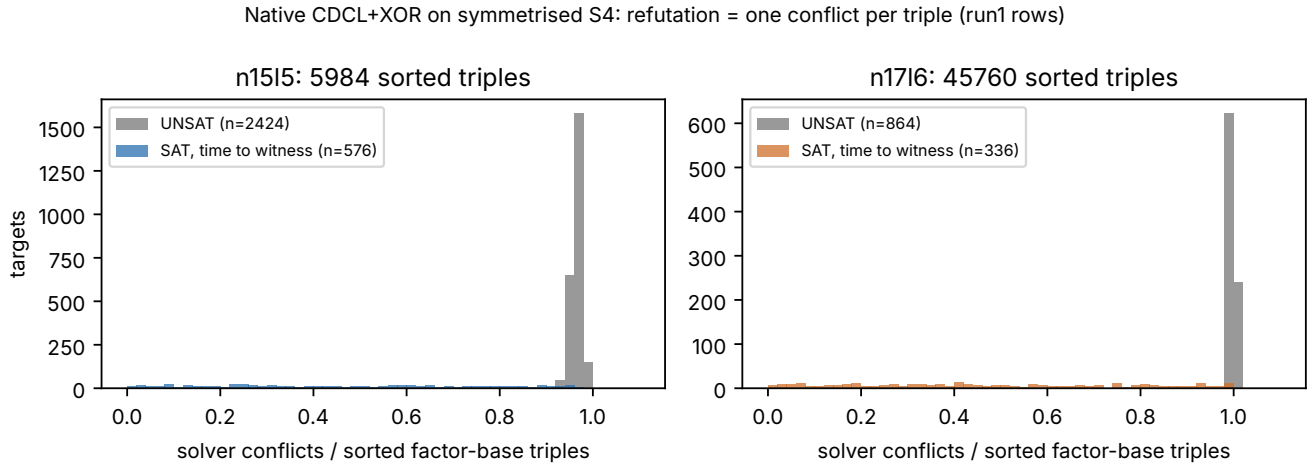



Figure 1: Solver conflicts per target over sorted triples (run1 rows, all 4200 targets).

**L1 triage (measured; gain derived as held-out relations/conflict ÷ solve-all, telemetry conflicts charged; CIs from 1000 bootstrap resamples).**

| family | split | features | dims | test AUC [95% CI] | permuted-label AUC | test gain [95% CI] | train gain | exact-label ceiling |
|---|---|---|---|---|---|---|---|---|
| n15l5 | even_odd | F-x | 15 | 0.515 [0.478, 0.552] | 0.510 | 0.79 [0.37, 1.31] | 1.99 | 9.01 |
| n15l5 | even_odd | F-xx | 120 | 0.503 [0.466, 0.539] | 0.495 | 1.11 [0.62, 1.77] | 3.49 | 9.01 |
| n15l5 | even_odd | F-tel64 | 11 | 0.514 [0.476, 0.551] | 0.492 | 1.08 [0.85, 1.35] | 1.27 | 8.28 |
| n15l5 | even_odd | F-tel512 | 11 | 0.556 [0.518, 0.599] | 0.508 | 0.99 [0.85, 1.13] | 1.14 | 5.27 |
| n15l5 | aa_mod4 | F-x | 15 | 0.526 [0.491, 0.566] | 0.537 | 0.98 [0.55, 1.43] | 1.79 | 8.83 |
| n15l5 | aa_mod4 | F-xx | 120 | 0.485 [0.450, 0.523] | 0.540 | 1.39 [0.91, 1.96] | 3.95 | 8.83 |
| n15l5 | aa_mod4 | F-tel64 | 11 | 0.531 [0.494, 0.567] | 0.534 | 1.17 [0.92, 1.44] | 1.30 | 8.12 |
| n15l5 | aa_mod4 | F-tel512 | 11 | 0.540 [0.501, 0.575] | 0.478 | 0.91 [0.73, 1.09] | 1.25 | 5.21 |
| n17l6 | even_odd | F-x | 17 | 0.496 [0.444, 0.547] | 0.517 | 0.65 [0.19, 1.26] | 1.69 | 6.69 |
| n17l6 | even_odd | F-xx | 153 | 0.461 [0.408, 0.514] | 0.471 | 0.89 [0.29, 1.68] | 4.67 | 6.69 |
| n17l6 | even_odd | F-tel64 | 11 | 0.508 [0.460, 0.561] | 0.502 | 1.19 [0.33, 2.56] | 1.39 | 6.64 |
| n17l6 | even_odd | F-tel512 | 11 | 0.513 [0.462, 0.569] | 0.499 | 1.30 [0.69, 2.05] | 1.59 | 6.29 |
| n17l6 | aa_mod4 | F-x | 17 | 0.488 [0.437, 0.541] | 0.480 | 0.63 [0.14, 1.32] | 2.68 | 6.08 |
| n17l6 | aa_mod4 | F-xx | 153 | 0.484 [0.435, 0.536] | 0.462 | 1.27 [0.57, 2.08] | 4.04 | 6.08 |
| n17l6 | aa_mod4 | F-tel64 | 11 | 0.486 [0.438, 0.541] | 0.449 | 0.78 [0.23, 1.57] | 1.68 | 6.03 |
| n17l6 | aa_mod4 | F-tel512 | 11 | 0.505 [0.457, 0.557] | 0.486 | 0.94 [0.70, 1.19] | 1.41 | 5.75 |

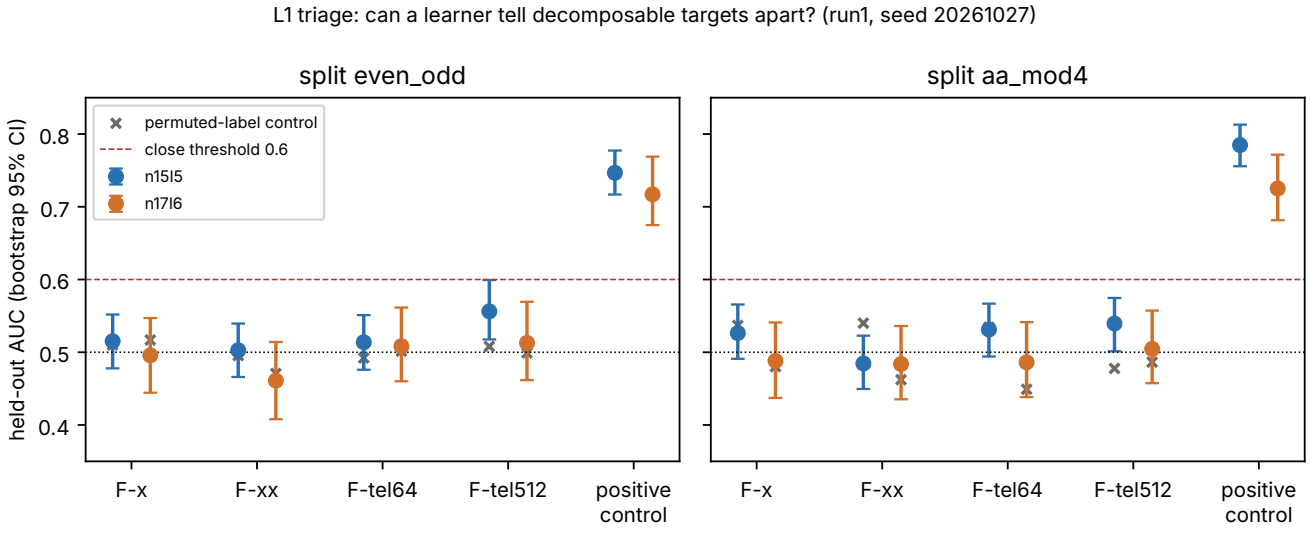



Figure 2: Held-out AUC per feature group with positive and permuted-label controls (run1 analysis).

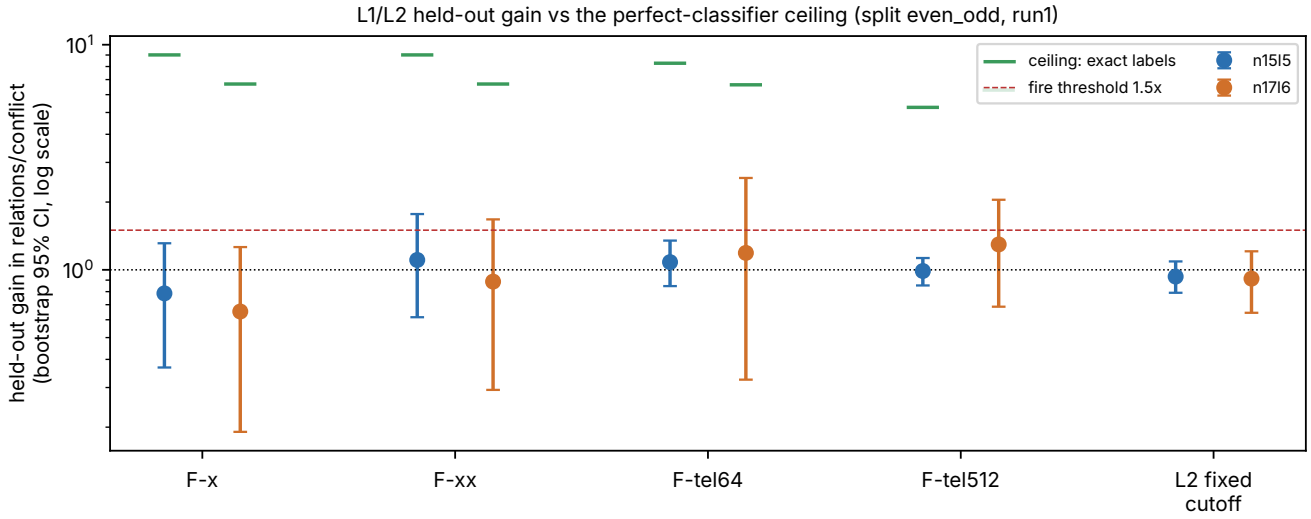



Figure 3: Held-out gain against the exact-label ceiling, split even_odd (run1 analysis).

**L2 fixed cutoff and the positive control.**

| family | split | train-optimal cutoff (conflicts) | train gain | test gain [95% CI] | positive-control AUC | positive-control gain |
|---|---|---|---|---|---|---|
| n15l5 | even_odd | 1922 | 1.25 | 0.93 [0.79, 1.09] | 0.747 [0.717, 0.777] | 2.11 |
| n15l5 | aa_mod4 | 1915 | 1.12 | 1.05 [0.91, 1.19] | 0.785 [0.756, 0.813] | 2.81 |
| n17l6 | even_odd | 9022 | 1.12 | 0.91 [0.64, 1.21] | 0.717 [0.675, 0.769] | 2.23 |
| n17l6 | aa_mod4 | 3264 | 1.33 | 0.98 [0.50, 1.55] | 0.725 [0.682, 0.772] | 2.74 |

**Size transfer (telemetry model trained on n15l5, scored on n17l6).**

| features | trained on | scored on | AUC [95% CI] |
|---|---|---|---|
| F-tel64 | n15l5 (all 3000) | n17l6 (all 1200) | 0.496 [0.458, 0.531] |
| F-tel512 | n15l5 (all 3000) | n17l6 (all 1200) | 0.520 [0.482, 0.557] |

**L3 branching order (capped arms: values are lower bounds).**

| family | arm | targets | capped | median conflicts | sum / a0 | conflicts per sorted triple | conflicts per pair |
|---|---|---|---|---|---|---|---|
| n15l5 | a0_default | 60 | 0 | 5759 | 1.000 | 0.963 | 5.629 |
| n15l5 | a1_x_all | 60 | 0 | 5759 | 1.000 | 0.963 | 5.629 |
| n15l5 | a2_x1x2 | 60 | 0 | 6466 | 1.120 | 1.079 | 6.304 |
| n15l5 | a3_e_vars | 60 | 60 | ≥ 57810 | ≥ 10.030 | ≥ 9.661 | ≥ 56.455 |
| n15l5 | a4_learned_activity | 60 | 60 | ≥ 57810 | ≥ 10.030 | ≥ 9.661 | ≥ 56.455 |
| n17l6 | a0_default | 30 | 0 | 45600 | 1.000 | 0.998 | 11.145 |
| n17l6 | a1_x_all | 30 | 0 | 45600 | 1.000 | 0.998 | 11.145 |
| n17l6 | a2_x1x2 | 30 | 0 | 55806 | 1.220 | 1.217 | 13.599 |
| n17l6 | a3_e_vars | 30 | 30 | ≥ 456490 | ≥ 10.000 | ≥ 9.976 | ≥ 111.448 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| n17l6 | a4_learned_activity | 30 | 30 | ≥ 456490 | ≥ 10.000 | ≥ 9.976 | ≥ 111.448 |

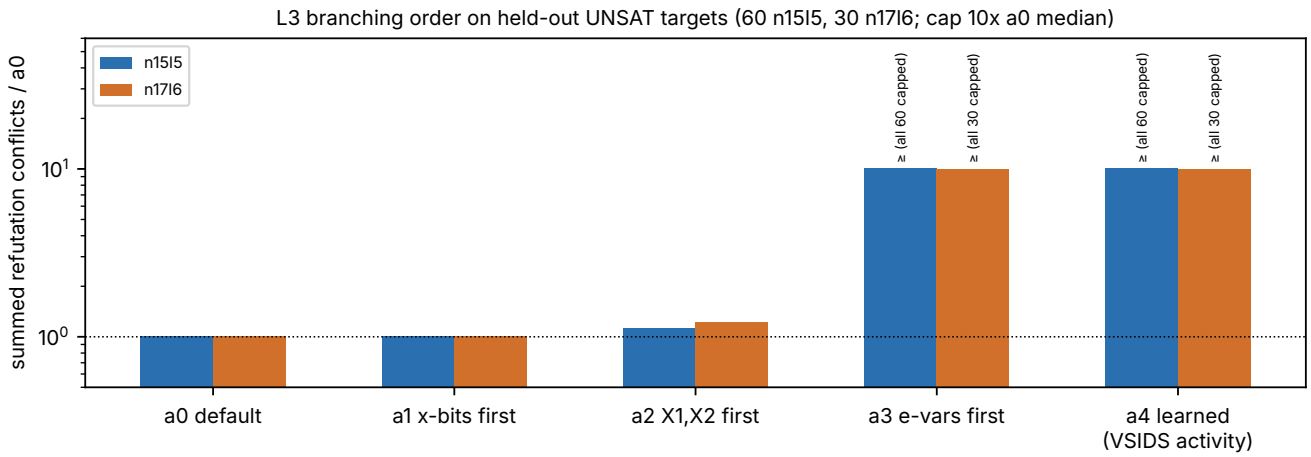



Figure 4: Summed refutation conflicts relative to default VSIDS (run1 analysis).

**Raw records (SHA-256), paths under `run1/`.**

`rows_n15l5.jsonl` 40f5fee112f8dd271dddd3477fa2a02bc29a8bc250c6f975a0149b0ca8b81eec
`rows_n17l6.jsonl` b959864d7064a3965a2e4255e51c0b40378f1a942adbf74f79cf54d4ce188673
`analysis/l3_n15l5.json` f789f371a7e5e440b1204e69247811d029e93fa104fd305e126bfeb87f5dfc4f
`analysis/l3_n17l6.json` 5e731b6bd568ab7e1c2ce521903872263f388b180c7de45e0634d0b3bf370403
`analysis/summary_n15l5.json` 9df7be93e8317c742915a2e55da44ac1699ddb378ec95eaa6f1b459c41f1e31e
`analysis/summary_n17l6.json` a273bacdd546c5938337ea1f529c0722cf13237e64f15f1f4667e52ff2cd33d1
`analysis/transfer.json` 6ba8f3fdf66110f79a990a1cde8a7c32b7ff5ad4b627c2947f970aeb59705626

Observations only; the interpretation and the verdict under the frozen rules are in the study README.